

- 4 (a) Molecule has component of velocity in three directions

$$c^2 = c_x^2 + c_y^2 + c_z^2$$

B1

$$\text{Averaging: } \langle c^2 \rangle = \langle c_x^2 \rangle + \langle c_y^2 \rangle + \langle c_z^2 \rangle$$

B1

$$\text{Due to random motion: } \langle c_x^2 \rangle = \langle c_y^2 \rangle = \langle c_z^2 \rangle$$

B1

$$\langle c^2 \rangle = 3 \langle c_x^2 \rangle$$

$$\langle c_x^2 \rangle = \frac{1}{3} \langle c^2 \rangle$$

$$P = \frac{Nm \langle c_x^2 \rangle}{V} = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

A0

(b) work done = $p\Delta V = 5.7 \times 10^6 \text{ Pa} \times (3.1 - 2.0) \times 10^{-5} \text{ m}^3$

1

$$= 62.7 \text{ J}$$

1

(c)

section of cycle	heat supplied to the gas / J	work done on the gas / J	increase in the internal energy of the system / J
A → B	0	235	235 A1
B → C	246	− 63 B1	183 B2 (sum of 246 and −63)
C → D	0	− 333	− 333 A2
D → A	−85 C2	0 C1	−(235 + 183 − 333) = − 85 C2

A1 & A2 (1), B1 & B2 (1), C1 & C2 (1), no ECF

Comments: In (a), many mixed up $\langle c_x^2 \rangle$ and $\langle c_x \rangle^2$, as well as the expressions $\langle c_x^2 \rangle = \langle c_y^2 \rangle = \langle c_z^2 \rangle$ and $c_x = c_y = c_z$. In (b), some gave negative value for the work done by the gas. Most answered part (c) well, though a few were careless in additions and subtractions when using $\Delta U = Q + W$ and $\Delta U_{\text{cyclic process}} = 0$.

5 (a) (i) Waves from loudspeaker travel down tube and are reflected at closed end